So, if not protons and neutrons, what constitutes the dark matter? As of today, no one knows, but there is no shortage of proposals. The candidates' names run the gamut from axions to zinos, and whoever finds the answer will surely pay a visit to Stockholm. That no one has yet detected a dark matter particle places significant constraints on any proposal. The reason is that dark matter is not only situated out in space; it is distributed throughout the universe and so is also wafting by us here on earth. According to many of the proposals, right now billions of dark matter particles are shooting through your body every second, so viable candidates are only those particles that can pass through bulky matter without leaving a significant trace.

Neutrinos are one possibility. Calculations estimate their relic abundance since they were produced in the big bang, at about 55 million per cubic meter of space, so if any one of the three neutrino species weighed about a hundredth of a millionth (10^{-8}) as much as a proton, they would supply the dark matter. Although recent experiments have given strong evidence that neutrinos do have mass, according to current data they are too light to supply the dark matter; they fall short of the mark by a factor of more than a hundred.

Another promising proposal involves supersymmetric particles, especially the *photino*, the *zino*, and the *higgsino* (the partners of the photon, the Z, and the Higgs). These are the most standoffish of the supersymmetric particles—they could nonchalantly pass through the entire earth without the slightest effect on their motion—and hence could easily have escaped detection.[9] From calculations of how many of these particles would have been produced in the big bang and survived until today, physicists estimate that they would need to have mass on the order of 100 to 1,000 times that of the proton to supply the dark matter. This is an intriguing number, because various studies of supersymmetric-particle models as well as of superstring theory have arrived at the same mass range for these particles, without any concern for dark matter or cosmology. This would be a puzzling and completely unexplained confluence, unless, of course, the dark matter is indeed composed of supersymmetric particles. Thus, the search for supersymmetric particles at the world's current and pending accelerators may also be viewed as searches for the heavily favored dark matter candidates.